Practice Exam Solutions — Lecture 1–3

Phy 4143 · Full worked solutions to practice-exam-lecture1-3.pdf, same question numbering.

Q1. Theory and Definitions — model answers

a) Voltage, Current, Ohm's Law

Voltage: electrical potential difference between two points; the force driving current, in volts (V). Current: the flow of electric charge through a conductor, in amperes (A). Ohm's Law: V = I × R — voltage is directly proportional to current and inversely proportional to resistance.

b) Electric circuit, KCL, KVL

Electric circuit: a closed path for transmitting current through electrical and magnetic fields. KCL: the algebraic sum of currents at any node is zero (current in = current out). KVL: the algebraic sum of voltages around any closed loop is zero.

c) Node, junction, branch, loop, mesh

Node: a point where two or more elements join. Junction: a point where three or more elements meet (current actually divides here — this is where KCL is meaningful). Branch: the part of a network between two junction points. Loop: any closed path. Mesh: the smallest loop that can't be divided further. **Node vs junction:** every junction is a node, but a node with only 2 elements meeting (no current split) is not called a junction.

d) Ideal voltage source vs ideal current source

Voltage source: a two-terminal device whose voltage stays constant no matter what current is drawn; ideal internal resistance = 0 Ω (zero series resistance). Current source: supplies the same current regardless of the voltage across it; ideal internal resistance = ∞ (infinite, i.e. an open circuit internally).

e) Source transformation

Valid only when converting a *practical* source (one with a series/parallel internal resistance R) — an ideal source alone cannot be transformed. Voltage → Current: Is = V/R, with R moved from series to parallel. Current → Voltage: V = Is × R, with R moved from parallel to series. The terminal V-I behaviour (as seen by the rest of the circuit) is identical before and after.

f) Series and parallel combination

Series (same current through every element): Req = R1 + R2 + … + Rn. Parallel (same voltage across every element), for two resistors: Req = (R1×R2)/(R1+R2); in general 1/Req = 1/R1 + 1/R2 + … + 1/Rn.

g) Voltage Divider Rule and Current Divider Rule

Voltage Divider: Vx = (Rx/Rs) × V, where Rx is the resistor you want the voltage across and Rs is the total series resistance of the string. Current Divider: Ix = (Rp/Rx) × I, where Rp is the equivalent resistance of the whole parallel combination and Rx is the branch whose current you want.

h) When Y-Δ transformation is needed

When a resistor network contains a “bridge” — three resistors meeting at an internal node that cannot be identified as purely series or purely parallel with the rest of the circuit (e.g. a Wheatstone-bridge-shaped network) — plain series/parallel reduction gets stuck, and converting the Y (or Δ) at that internal node lets the reduction continue.

Q2. Basic Laws — full solutions

a) 30 V, series 2 Ω + 3 Ω + 5 Ω

Req = 2+3+5 = 10 Ω. i = 30/10 = 3 A. v1=3×2=6 V, v2=3×3=9 V, v3=3×5=15 V. KVL check: 6+9+15 = 30 V, matches the source.

Answer: i = 3 A, v1 = 6 V, v2 = 9 V, v3 = 15 V

b) 18 V & 6 V aiding, series 3 Ω + 5 Ω

Aiding sources add: Vtotal = 18+6 = 24 V. Req = 3+5 = 8 Ω. i = 24/8 = 3 A. v(3Ω) = 3×3 = 9 V, v(5Ω) = 3×5 = 15 V.

Answer: i = 3 A, v(3Ω) = 9 V, v(5Ω) = 15 V

c) Single-node KCL — 8 A, 4 Ω (i_o), dependent $i_o/2$, 8 Ω (v_o)

Since the 4 Ω and 8 Ω share the same node, $v_o = i_o \times 4$ (Ohm's law on the 4 Ω branch). KCL at the node (current in = current out): $8 = i_o + i_o/2 + v_o/8$. Substitute $v_o = 4i_o$, so $v_o/8 = i_o/2$: $8 = i_o + i_o/2 + i_o/2 = 2i_o$.

Answer: $i_o = 4$ A, $v_o = 4 \times 4 = 16$ V

d) Source transformation — 20 V + 5 Ω, feeding a 15 Ω load

(i) Is = V/R = 20/5 = 4 A, with the same 5 Ω now in parallel. (ii) Current divider between the source's own 5 Ω and the 15 Ω load: i(load) = [5/(5+15)] × 4 = 1 A. v(load) = i × R = 1×15 = 15 V.

Answer: (i) Is = 4 A with 5 Ω in parallel (ii) i(load) = 1 A, v(load) = 15 V

Q3. Connections, Dividers, Y-Δ — full solutions

a) R_eq between A and B — 4 Ω series (6 Ω parallel 3 Ω) series 5 Ω

6 parallel 3 = (6×3)/(6+3) = 18/9 = 2 Ω. R_eq = 4+2+5 = 11 Ω.

Answer: R_eq = 11 Ω

b) Voltage Divider — R1=4, R2=6, R3=10 Ω, 40 V series

v2 = [R2/(R1+R2+R3)] × V = [6/20] × 40 = 12 V.

Answer: v2 = 12 V

c) Current Divider — R1=6, R2=12, R3=4 Ω parallel, 18 A total

1/Rp = 1/6+1/12+1/4 = 2/12+1/12+3/12 = 6/12 = 1/2 → Rp = 2 Ω. i1 = (Rp/R1) × I = (2/6) × 18 = 6 A.

Answer: i1 = 6 A

d) Y → Δ — R1=4, R2=6, R3=12 Ω (star point to a, b, c)

sum = R1R2+R2R3+R3R1 = 24+72+48 = 144. R_A = sum/R1 = 144/4 = 36 Ω. R_B = sum/R2 = 144/6 = 24 Ω. R_C = sum/R3 = 144/12 = 12 Ω.

Answer: R_A = 36 Ω, R_B = 24 Ω, R_C = 12 Ω

e) Δ → Y verification, using R_A=36, R_B=24, R_C=12 Ω

Δ→Y formula: each Y arm = (product of the two Δ sides touching that arm's node) divided by the simple sum R_A+R_B+R_C = 36+24+12 = 72. R1' = (R_B×R_C)/72 = (24×12)/72 = 288/72 = 4 Ω. R2' = (R_C×R_A)/72 = (12×36)/72 = 432/72 = 6 Ω. R3' = (R_A×R_B)/72 = (36×24)/72 = 864/72 = 12 Ω.

Answer: R1 = 4 Ω, R2 = 6 Ω, R3 = 12 Ω — matches the original Y values exactly